

Experiment No. 7

Inorganic Qualitative Analysis (Salt no. 3)

Aim: Analyse one basic (cation) and one acidic (anion) radical from given inorganic salt.

Apparatus: Test tubes, test tube stand, spatula, glass rod, watch glass, beaker, burner, etc.

A. Preliminary tests: (General scheme)

Test	Observation	Inference
Nature	Crystalline (soluble in water)	Cl^- , Br^- , I^- , Hg_2^{2+} , SO_4^{2-} , may be Present
Color of salt	White	Pb^{2+} , Al^{3+} , Mg^{2+} , Ba^{2+} , Zn^{2+} , NH_4^+ , Ca^{2+} , Mn^{2+} , Sr^{2+} may be Present
Solubility	Soluble in water	Crystalline salt Present.

B. Dry tests for basic radicals (Cations)

Test	Observation	Inference
1. Heating in a dry test tube:- Take a small quantity of salt in a clean, dry test tube and heat it strongly. Observe the color of the residue and gas evolved.	Salt fuses	Salts of Ca^{2+} and hydrated salts may be Present
2. Test for NH_4^+ Salt + NaOH, heat gently, hold moist turmeric paper near the mouth of test tube.	No reaction	—
3. Charcoal cavity test: Take the salt and solid Na_2CO_3 in the ratio 1:2 on watch glass, add few drops of water and prepare a paste, place this in charcoal cavity and heat it with a blow pipe in a reducing flame.	White infusible residue	Zn^{2+} , Al^{3+} , Mg^{2+} , Ba^{2+} , Sr^{2+} , Ca^{2+} , may be Present
4. Cobalt Nitrate Test: If white infusible residue is obtained in charcoal cavity test, add two drops of cobalt nitrate solution on the white residue and heat in the blue flame.	Residue turns grey	Ba^{2+} , Ca^{2+} , may be Present.

5. Flame test: Clean a platinum wire with conc. HCl. Moisten a small quantity of the substance with concentrated HCl on watch glass. Dip the cleaned platinum wire in it and heat it in oxidizing flame (non - luminous) and observe the color of the flame.	Brick red	Ca^{2+} may be Present
6. Borax bead test: [For colored salt only] Make a small loop at the end of a clean platinum wire heat it strongly and dip it in borax powder. Heat it again till a glossy bead is obtained. Now touch the hot bead with a small crystal of the salt and heat it again in oxidizing (blue) flame. Observe the color of the bead.	—	—

C. Dry tests for acidic radicals (Anions)

Test	Observation	Inference
i. Action of dil HCl: Salt + dil HCl.	No reaction	—
ii. Action of conc. H_2SO_4: Salt + conc. H_2SO_4 + heat	colourless gas giving white dense fumes glass rod dipped in NaOH	Cl^- may be Present
iii. Action of MnO_2 and conc. H_2SO_4: Salt + a pinch of MnO_2 + conc. H_2SO_4 and heat	Greenish yellow gas turning moist blue litmus paper red and then bleaches.	Cl^- may be Present.
iv. Cu-foil test: (If brown gas is obtained in the above test then only perform this test) Salt + Cu-foil + conc. H_2SO_4 and heat strongly	—	—

Note: (If the anion is not detected in above dry test, then salt may contain SO_4^{2-} or PO_4^{3-})

D. Preparation of solution (Original Solution): Dissolve about 0.5 gm of salt in 20 mL of water in a 50 mL beaker. This is called original solution (O.S.)

E. Wet test for basic radical (cations)

Detection of groups

Test	Observation	Inference
1. O.S. + dil. HCl	No ppt	Group I absent
2. O.S. + dil. HCl + H ₂ S gas / H ₂ S water solution	No ppt	Group II absent
3. O.S. + NH ₄ Cl + NH ₄ OH till alkaline	No ppt	Group III A absent
4. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + H ₂ S gas / H ₂ S water solution	No ppt	Group III B absent
5. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + saturated solution of (NH ₄) ₂ CO ₃	white ppt	Group IV Present
6. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + Na ₂ HPO ₄	No reaction	Group V absent
7. If all the above groups are absent	-	-

Analysis of Group: After detection of a particular group do analysis of that group.

Analysis of Group IV

Test	Observation	Inference
0.5 + K ₂ CrO ₄	No ppt	Sr ²⁺ or Ca ²⁺
C.T. for..... 1. 0.5 + saturated solution of (NH ₄) ₂ SO ₄ and boil	No ppt	Ca ²⁺ confirmed
2. 0.5 + K ₄ [Fe(CN) ₆]	white ppt	Ca ²⁺ confirmed

F. Wet test for acidic radicals:

Detection of acidic radical from water solution:

1. Wet tests for CO_3^{2-} , SO_3^{2-} , Cl^- , Br^- , I^-

1. O.S. + AgNO_3	Yellow ppt insoluble in dil. HNO_3	Cl^- , Br^- , I^- are present
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Distinction between halides (Cl^- , Br^- , I^-):

Note: (Perform this test only if ppt is obtained for AgNO_3 which is insoluble in dil. HNO_3)

Test	Observation	Inference
O.S. + Chloroform + Cl_2 water in excess (shake well)	chloroform layer is colourless	Cl^- present

Note: (If one of the halide radical is present then further analysis is not necessary. Perform confirmatory test for that detected radical.)

2. Wet test for SO_4^{2-} and PO_4^{3-}

(This test is required to perform only, if ppt is NOT obtained in AgNO_3 test)

Test	Observation	Inference
O.S. + $\text{Ba}(\text{NO}_3)_2$	_____	_____

3. WET TEST FOR NO_2^-

Test	Observation	Inference
O.S. + dil. H_2SO_4 + FeSO_4 solution	_____	_____

4. WET TEST FOR NO_3^-

Test	Observation	Inference
O.S. + 3-4 drops of conc. H_2SO_4 + 2-3 drops of diphenyl amine	_____	_____

5. WET TEST FOR CH_3COO^-

Test	Observation	Inference
O.S. + FeCl_3 solution	_____	_____

6. WET TEST FOR $\text{C}_2\text{O}_4^{2-}$

Test	Observation	Inference
O.S. + dil. Acetic acid + CaCl_2		

Cl

C.T. for

Test	Observation	Inference
1. O.S + MnO_2 + 2-3 drops of conc. H_2SO_4 & heat	Greenish yellow gas evolves that turns moist blue litmus red & bleaches it	Cl^- confirmed
2. O.S + lead acetate	White ppt	Cl^- confirmed.

Result:- Given inorganic salt no. 3 contains following

- Basic radical Ca^{2+} (Calcium ion)
- Acidic radical..... Cl (Chloride ion)
- Molecular formula of the salt..... CaCl
- Hence the name of the salt..... Calcium chloride

Remark and sign of teacher: